SensoSpoon: Design and Simulation-Based Validation of a Two-Axis Active Tremor-Stabilization Eating Device

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***Abstract*—Hand tremor is a common and highly disruptive symptom of essential tremor (ET) and Parkinsonian tremor that substantially impairs independent eating. This paper presents the mechanical, electronic and control design of SensoSpoon, a handheld assistive eating utensil that actively stabilizes spoon attitude using a two-axis flexure gimbal driven by two voice-coil actuators, informed by dual inertial measurement units (IMUs) and a 13-state extended Kalman filter (EKF) that separates voluntary motion from tremor, gated by surface electromyography (sEMG). A structured review of six tremor-suppression precedents and an end-user interview motivated the architecture and its two dominant usability constraints: minimal resistance to intended motion, and low weight, bulk and cost. A 20-scenario controlled benchmark, an sEMG ablation study, and validation against 60 recorded Parkinsonian tremor traces from the public PhysioNet tremordb database show that the delivered EKF+sEMG controller reaches 93.04% mean tremor suppression on synthetic tremor and 86.81% on recorded human tremor, against 71–73% for an IMU-only estimator and 82–90% for the original fixed-notch controller, while holding voluntary-motion tracking error to 0.207° versus 0.458° for the original design. These results support the technical feasibility of the proposed architecture under simulated and semi-realistic conditions; no physical prototype, bench measurement or human-subject recording was made for this project. Physical prototyping and closed-loop human validation remain necessary before any hardware or clinical performance claim can be made.**

Keywords— tremor suppression, assistive eating device, active stabilization, flexure gimbal, voice-coil actuator, inertial measurement unit, electromyography, extended Kalman filter, sensor fusion.

I. Introduction

Essential tremor and Parkinsonian tremor are among the most common movement disorders, and involuntary oscillation of the hand at typical frequencies of 4–12 Hz is one of their most functionally disabling symptoms during daily living [1]. Eating is particularly affected: a person who can grip an object may still be unable to carry a loaded spoon to the mouth without spilling, since tremor amplitude is often largest during active limb movement rather than at rest [2]. Adaptive and actively stabilized utensils range from simple weighted or ball-jointed spoons to sensor-driven, motor-actuated platforms, yet a 2025 comparative usability study found a simple weighted spoon outperformed several more complex actuated and counterweighted commercial devices on both food-transfer success and preference [3] — added mechatronic complexity does not automatically improve outcomes.

The central challenge is therefore not to oppose hand motion outright — resisting all movement would also resist the user's intended path to the mouth — but to distinguish tremor from voluntary motion and correct only the former, requiring both a credible estimate of intent and an actuator capable of rapid, bounded corrective torque.

This paper describes SensoSpoon, a handheld utensil that combines a flexure-based two-axis gimbal, two handle-mounted voice-coil actuators, dual IMUs and an sEMG-gated state estimator to perform bounded pitch and roll correction. The contribution is fourfold: (1) a mechanical/control architecture combining catalogue flexure pivots with a dual voice-coil pushrod drive; (2) end-user interview findings incorporated into the design requirements; (3) a 13-state EKF separating voluntary intent from tremor via a shared, estimated tremor frequency rather than a fixed filter band, gated by a signed sEMG agonist/antagonist confidence signal; and (4) quantification of that architecture's benefit and limits through a 20-scenario benchmark, an sEMG ablation, and a feasibility study replaying 60 recorded Parkinsonian tremor traces from a public database.

The remainder of the paper is organized as follows. Section II reviews prior tremor-suppression architectures and reports findings from a structured end-user interview. Section III describes the mechanical design, actuator, kinematics and motion behaviour of the gimbal. Section IV describes the sensing and the two-rate control cascade, including the EKF and sEMG intent gate. Section V presents the simulation and semi-realistic validation results. Section VI discusses these results and concludes.

II. Literature Review and End-User Interview

A. Architectural Precedent

Zartika et al. [4] built a handheld spoon whose IMU-based tremor estimate is refined by a Kalman filter, reducing pitch-angle error from 1.028° to 0.037° before driving a servo-actuated 2-DOF gimbal under PID control tuned by particle swarm optimization (PSO), with a measured step response of 0.09–0.34 s rise and 0.59–0.74 s settling — a concrete hardware benchmark for SensoSpoon's own loop.

The FiMec device [5] generalizes this with two DC motors on orthogonal handle axes under PID/genetic-algorithm control from a 3-axis IMU. Exercised in SimMechanics against recorded vibration data from ten essential-tremor patients, it reports 84–99.4% vibration absorption (average 93.47%) — a simulation ceiling, but confirmation that a handle-mounted, dual-motor, IMU–PID architecture is credible for this problem.

B. Clinical Validation

Pathak et al. [2], behind the commercial Liftware Steady device, provides the only published human-subject outcome reviewed. An accelerometer-driven 2-DOF platform tested on 15 ET patients reduced tremor amplitude 73% (Fahn–Tolosa–Marin scale), with significant improvement for eating ($p=0.001$) and transferring ($p=0.013$) but not passive holding ($p=0.14$) — active correction earns its benefit specifically during active manipulation, SensoSpoon's use case.

C. Broader Actuator Context

This subsection looks beyond eating utensils to tremor-suppression actuation in general, to situate SensoSpoon's actuator choice against the wider field rather than spoon-specific devices. Mo and Priefer's field-wide review [1] situates motor-driven actuation among competitors: a permanent-magnet linear motor reaches 97.6% suppression, an EC-motor elbow orthosis [6] 99.8%, while a magnetorheological-damper orthosis is listed "not reported." These figures come from wrist- and elbow-worn orthoses, not spoons. Among feeding devices specifically, the Gyenno Spoon claims 85% suppression, but unlike Liftware has no peer-reviewed clinical support.

D. Control-Law Precedent

Taheri et al. [7] designed a pneumatic 1-DOF wrist orthosis with a high-pass tremor estimator and a backstepping law with a full Lyapunov proof, stress-tested against 100–133% stiffness/damping variation. On real patient data it achieved 30.3–41.9 dB (97.5–99.2%) suppression, and characterized tremor as a 4–6 Hz fundamental with an 8–12 Hz harmonic carrying 94.7–96.4% of tremor energy — SensoSpoon's original target band, subsequently widened to a 2–25 Hz search once the frequency-tracking EKF removed the need for a fixed band (Section IV-B2).

Herrnstadt and Menon [6] instead estimate only the voluntary component and track it with an admittance/velocity controller, treating tremor as a rejected disturbance rather than a signal to invert: >99% power reduction with only -0.15% voluntary distortion, close in spirit to the intent/tremor-separating EKF ultimately adopted here (Section IV-B2).

E. Summary of Evidence

Table I consolidates the architecture, actuation strategy and quantitative result of each source reviewed.

Source	Architecture	Result
Zartika et al. [4]	2-DOF servo gimbal, IMU+Kalman, PID/PSO	0.09–0.34 s rise, 0.59–0.74 s settling (hardware)
FiMec [5]	2-DOF DC-motor gimbal, 3-axis IMU, PID/GA	84–99.4% (avg 93.47%) absorption (SimMechanics)
Pathak et al. [2]	2-DOF motor platform, accelerometer	73% reduction; 15 ET patients, 3 tasks (clinical)
Mo & Priefer [1]	Field survey: motors, MR/EM brakes, pneumatics	PMLM 97.6%, EC motor 99.8%, pneumatic 98.1%, MR n.r.
Taheri et al. [7]	1-DOF pneumatic wrist orthosis, backstepping	30.3–41.9 dB (97.5–99.2%); robust to 100–133% Δ
Herrnstadt & Menon [6]	1-DOF EC-motor elbow orthosis, voluntary-tracking	>99% suppression; −0.15% voluntary distortion

TABLE I. Summary of Reviewed Sources

F. Synthesis and Design Implications

Collectively, these sources build a defensible case for a motor-driven 2-DOF gimbal, with the strongest clinical evidence confirming benefit during active manipulation and two independently validated control laws directly relevant once the mechanical design is fixed. Three gaps were apparent at the outset: no reviewed source reported closed-loop bandwidth against the tremor band [7]; no reviewed source compared multiple estimator/control architectures under an identical scenario set, each having validated only its own single design; and none of the six precedents paired sEMG with the IMU for sensing or suppression — every reviewed architecture estimates tremor from inertial or accelerometer data alone. Sections IV and V report how far this work has closed all three: analytical verification of closed-loop bandwidth (Section III-C), a six-variant benchmark under identical conditions (Section V-D), and an sEMG ablation isolating what pairing it with the IMU contributes (Section V-E).

G. End-User Interview

A structured interview was conducted with a person experiencing hand tremor to ground the design requirements. The participant reported tremor both at rest and during movement, and identified holding the spoon steady as the hardest part of eating; the same tremor also affects other light-object tasks such as writing. Their only coping strategy is gripping the spoon tightly, and they had not previously tried a stabilizing spoon. Asked what such a spoon would need to do, they prioritized ease of carrying and, when asked directly, rated non-resistance to normal voluntary movement as "extremely important." A bulky or heavy appearance and high cost were the main reasons they would decline to use one; their ideal design was lightweight, portable, stylish and reasonably priced.

These responses map directly onto SensoSpoon's control philosophy: a bounded, disturbance-rejecting correction rather than one that opposes all handle motion, and a compact, handle-integrated dual-actuator layout. The interview's emphasis on non-resistance during voluntary movement is revisited critically in Section VI in light of the fast-reach limitation identified by testing (Section V-F).

III. DESIGN

A. Mechanical Arrangement

SensoSpoon uses a two-axis flexure gimbal to adjust the spoon orientation relative to the handle. The handle houses the actuators and electronics, while the detachable spoon connects

to a moving carrier inside the spherical front cover (Fig. 1). Pitch rotates the bowl about a transverse axis, and roll tilts it about its longitudinal axis. Together, these motions allow the spoon to counter unwanted handle rotation as the user guides it along the feeding trajectory. The mechanism does not provide independent translational control.

Two C-Flex B-10 flexure pivots support pitch, and one C-Flex D-10 pivot supports roll. Rotation occurs through elastic deformation of their internal springs, which also produce a restoring torque that contributes to the actuator load [8], [9]. These pivots constrain the carrier to the two intended axes within the spherical cover.The device is 271 mm long. Its bowl measures 62 × 40 × 12 mm, with a 1.2 mm rim and a 1.8 mm bottom wall, and connects to the carrier through a 31 mm curved neck. The 34 mm moving carrier sits within a front bezel of 42 mm maximum diameter. Nominal joint travel is ±15° on each axis; commanded references are limited to ±14° to leave a margin before the compliant end stops. C-Flex specifies indefinite life at ±15° only at zero load [8], [9].



Fig. 1. Current MATLAB/CAD geometry: internal-arrangement cutaway showing the staggered actuator pair, gimbal carrier and main PCB.

B. Voice-Coil Actuation

Two Moticont GVCM-016-019-01M guided linear voice-coil motors [10] drive the carrier through direct spherical-ended pushrods. The actuator housings are fixed to the handle, while their moving coils and shafts translate. The articulated rod ends accommodate changes in alignment as the gimbal rotates. To fit within the handle, the two 15.9 mm diameter actuators are staggered along its length, with the pushrods terminating at a common neutral plane.

Axial force is approximately proportional to coil current, and reversing the current reverses the force. Each actuator has 12.7 mm total travel, or ±6.35 mm about its centred position. The catalogue force ratings are 1.5 N continuously and 4.6 N intermittently at 10% duty, with a specified shaft side-load limit of 2.2 N [10]. The model uses a 7.4 V supply and a 1.5 A current limit, with a nominal food payload of 5 g at the bowl. Table II lists the principal operating parameters.

C. Coupled Kinematics

Pushrod forces act at offset attachment points on the gimbal, generating moments about the pitch and roll axes. Both rods contribute to both rotations, and their moment arms vary with the joint configuration. The controller must therefore coordinate the two actuators through the linkage geometry. For joint coordinates $\mathbf{q} = [q_p, q_r]^{\mathrm{T}}$, the fixed rod-length constraints determine actuator displacements $\mathbf{s}(\mathbf{q})$. The transmission Jacobian is

$$\mathbf{J}(\mathbf{q}) = \frac{\partial \mathbf{s}}{\partial \mathbf{q}}.$$

It maps joint velocities to actuator velocities and actuator forces to joint torques:

$$\dot{\mathbf{s}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}},$$
$$\mathbf{Q} = \mathbf{J}^{\mathrm{T}}(\mathbf{q})\mathbf{F}.$$

Here, $\mathbf{F}$ contains the two axial actuator forces and $\mathbf{Q}$ contains the pitch and roll torques. The force mapping follows from virtual work; using the same Jacobian in both mappings preserves mechanical power [11]. Section V-B reports the rod-closure, Jacobian and power-balance checks used to verify numerical consistency.

D. Electromechanical Model

The plant has eight continuous states: two joint angles, two angular velocities, two coil currents and two coil temperature rises. Its mechanical dynamics include moving-assembly inertia, gravity and loads induced by handle motion, together with flexure restoring torque, damping and guide friction. Compliant end stops apply contact torques, allowing their effect on the response to be calculated without clipping the integrated joint angles.

Each coil follows an RL electrical model with back electromotive force and temperature-dependent resistance. A position-dependent force constant converts coil current into axial force. The same constant governs back electromotive force, preserving consistency between electrical and mechanical power. Applied voltage therefore determines current, actuator force and, through the coupled linkage, spoon motion.

Flexure hysteresis, creep and backlash are omitted, and guide friction is simplified. The thermal network, force variation with stroke, battery resistance and several mass properties remain estimates. Assembly clearances, shaft-extension bending and flexure fatigue under load also require experimental verification. The model supports controller development and comparison, with physical performance still to be established through prototyping and bench tests.

E. Material, Manufacturing, and Recyclability

The handle shell and the detachable spoon head are both molded in PCTG, a food-safe copolyester chosen over PP and PC for its high impact strength (>800 J/m) and chemical/dishwasher resistance without PC's brittleness risk; keeping both food-contact parts in one resin family also keeps end-of-life sorting to a single stream.

The handle grip is a separate, press-fit TPE (Shore A 50) sleeve rather than a bonded overmold, so it can be removed from the shell for recycling or replacement; its inner surface carries a molded-in herringbone texture angled toward the palm to resist axial slip, giving both compliant damping and mechanical grip for tremor-affected users. Pushrods, linkages and flexure pivots use aluminium (6061-T6/7075-T6) and catalogue 17-7 PH stainless steel rather than plastic, since the transmission Jacobian derived in Section III-C assumes near-rigid links; fasteners use brass heat-set threaded inserts.

All PCTG and TPE parts are sized for high-volume injection molding using tooling and shrinkage profiles compatible with standard PET/PETG processing; metal components are precision-machined or catalogue-sourced. PCTG carries resin code 1 (PET family); since the TPE grip sleeve is press-fit rather than chemically bonded, it separates from the shell by hand without shredding, keeping each resin in its own single-material recycling stream, while the mechanically fastened metal components remain fully scrap-recyclable at end-of-life.



Fig. 2. Final design render: PCTG handle shell and detachable spoon head with a press-fit, TPE grip sleeve.

IV. Electronics and Sensors

A. Sensing

Each IMU is sampled at 800 Hz and passed through a sensor model applying a one-sample output delay, a 100 Hz single-pole filter approximation, quantisation, range saturation, repeatable noise and a fixed unit-to-unit bias. Attitude is obtained by complementary filtering that relies principally on the gyroscopes, with slow accelerometer correction; because the accelerometers measure specific force including gravity and rotational acceleration, they do not report tilt directly, and large linear accelerations contaminate the gravity reference. Coil currents are measured through an independent 12-bit ADC path with its own delay, quantisation, noise and offset. Startup assumes two seconds of stationary, level dual-IMU calibration, which averages noisy samples rather than reading model bias constants.

B. Control Architecture

Control is a two-rate cascade. An inner 10 kHz current loop uses a PI regulator with resistance and back-EMF feedforward and back-calculation anti-windup on voltage saturation, so the integrator handles only model error. An outer 800 Hz attitude loop uses a PID law on carrier attitude error and measured angular rate, with mass-matrix, gravity, flexure and guide-friction compensation (ω_n=35 Hz, ζ=0.85); the resulting two-axis torque demand is mapped onto the two coils through the transmission Jacobian by a bounded least-squares allocator that enumerates the nine active-constraint cases directly, so no run-time optimisation toolbox is required. A supervisory layer (the sEMG intent gate and predictive authority limiter below) sits between the two loops, gating how much of the attitude error the inner loop is allowed to act on.

The design problem is not the servo loop. It is deciding what attitude to hold, because two requirements pull against each other: the bowl must stay level against 4–20 Hz involuntary motion, and it must follow the hand when the user deliberately moves. A filter aggressive enough to reject tremor also rejects the person.

1) Baseline - fixed notch chain: The original controller passes handle attitude through a fixed IIR chain — five notches at 4, 6, 8, 10 and 12 Hz with a 7 Hz bandwidth, bounded phase-lead compensation and a second-order 10 Hz low-pass, computed before startup with no online frequency search. It is simple, cheap and robust, but structurally cannot distinguish a slow voluntary movement from a low-frequency tremor, since the two overlap spectrally.

2) Intent/tremor separation by extended Kalman filter: The delivered controller replaces this with a thirteen-state EKF estimating voluntary and involuntary components separately: voluntary angle and rate per axis, tremor-fundamental and second-harmonic quadrature amplitudes, and a single tremor frequency ω shared across axes and estimated rather than gridded (2–25 Hz search, initialised at 6 Hz), so an out-of-band disturbance can still be tracked. The measurement is deliberately weak, $y=\theta_{intent}+a_1+a_2$ per axis: intent and tremor need only sum to the measurement, so tremor the single-frequency model cannot place is absorbed into the flexible intent state and passed to the spoon as though the user had meant it — undetectable from the IMU alone, which only ever sees the sum.

3) sEMG intent gate: The remedy is a second sensor observing the cause rather than the effect. Forearm agonist/antagonist sEMG is sampled at 4 kHz, band-passed 20–450 Hz, rectified and enveloped; the signed pair difference (isolating net joint torque from the tremor/effort-sensitive co-contraction floor) maps to a voluntary-drive confidence in [0,1] per axis that scales the EKF's intent process noise and does nothing else. Feeding a muscle-derived angle in as a pseudo-measurement was tried and rejected: it correlates well (r=0.95 – 0.98) but carries 1.6 – 1.8° of error against the 0.02 – 0.18° effect being removed. The gate runs a slow path (1 Hz corner, ≈225 ms delay) that may close it, and a fast path (4 Hz corner, ≈56 ms) that may only open it, combined with a maximum — wrongly closing suppresses the user's movement, wrongly opening merely reverts to prior behaviour, so the two errors are not treated equally. This rests on a 30–50 ms electromechanical delay (muscle precedes limb motion) taken from the literature [12], not measured here.

4) Predictive authority limiter: A supervisory limiter derates control authority when the tremor correction predicted 20 ms ahead would exceed the coil current limit or joint travel, applying an 85% margin and ramping authority at 5 s^{-1} rather than switching, so the system fails gracefully into reduced suppression instead of saturation or end-stop contact.

C. Principal Parameters

Tables II and III give the plant/actuator and controller parameters not already stated in the text; all parameters are centralised in two configuration files, so the baseline and delivered controller can be exchanged without editing anything else.

Quantity	Value
Force constant K	1.5 N/A (nominal)
Coil resistance / inductance	4.6 Ω / 0.8 mH
Bus voltage	7.4 V (2S LiPo)
Current limit	1.5 A
Continuous copper power	4.4 W
Actuator stroke	±6.35 mm
Nominal payload	5 g at the bowl

TABLE II. Plant and Actuator Parameters

Quantity	Value
Current loop	10 kHz PI, kp 0.8, ki 2000, anti-windup 1500
EKF states	13 (intent ×4, tremor ×8, shared ω)
EKF process noise	q_intent 10^{-4}, q_tremor 10^{-5}, q_ω 2×10^{-2}
EKF measurement noise	$(0.02°)^2$

TABLE III. Controller Parameters

V. Validation through simulation

A. Scope of the Evidence

The evidence in this section is of two kinds, kept apart because they support different claims. The first is algorithmic simulation evidence: a controlled comparison of controller variants over a fixed scenario set, in a simulated plant, driven by synthetic tremor. This establishes whether the control architecture works, and it is the evidence against which the delivered controller was designed and tuned. The second is public real-data evidence: analysis of recordings made from human subjects by other researchers and released publicly, used in two distinct ways — to test the sEMG assumptions the fusion rests on, and to replace the synthetic tremor input with a recorded one.

There is no hardware tier. No physical prototype was built, no bench measurement was taken, and no human subject was recorded for this project; every actuator, coil and flexure parameter is a manufacturer data-sheet value applied to a structure that was never manufactured. The claim this work is entitled to make is therefore bounded: the control algorithm was validated in simulation, and public real-data analysis was used both to test the sEMG assumptions it rests on and to drive the plant with recorded human tremor. Physical closed-loop human validation was outside the scope of this project.

B. Simulation Environment and its Verification

The plant integrates mechanism kinematics, actuator electrical dynamics, payload inertia and sensor noise at a 0.5 ms mechanical step, with the outer attitude loop at 800 Hz and the inner current loop at 10 kHz nested inside it. Three things support treating this model as a usable development baseline. First, the mechanism is verified analytically: rod-closure error is 3.5×10^{-18} m, the Jacobian error 4.5×10^{-12} and the energy-balance residual 3.7×10^{-12} W, so the kinematic and power bookkeeping are exact to numerical precision. Second, the fast fixed-step solver used for the sweeps is checked against a continuous-time Simulink reference on a nominal case, and the two agree to within a small residual in joint angle and suppression. Third, the model is covered by an automated suite of seventeen validation groups — mesh properties, packaging clearances, physics, hardware limits, control behaviour, the estimator, the sEMG front end, the tremor input path and the liquid model — which runs in approximately 80 s and must pass before any result is generated. What the model does not contain matters equally: there is no hysteresis, no creep and no backlash in the flexure, and no friction model beyond a simple guide term. The plant is therefore an algorithm-development baseline, not a validated digital twin, and absolute performance figures inherit that limitation.

C. Test Protocol and Metric Definitions

The benchmark comprises twenty scenarios spanning four questions: a tremor-frequency sweep at 4, 5, 7, 8, 9, 11 and 12 Hz; tremor waveform and amplitude (a mixed multi-frequency tremor, a chirp, an irregular tremor, and a 3° amplitude case); voluntary motion, both alone and superimposed on tremor; and plant robustness (no payload, 15 g payload, a 5.8 V bus in place of 7.4 V, and a degraded actuator with +20% coil resistance, -15% force constant and doubled guide friction). Tremor suppression is the percentage reduction in attitude RMS relative to a baseline in which the spoon is rigidly attached to the hand under the identical base input, scored from t=1.5 s to exclude the start-up transient; it is retained as negative rather than clipped when a controller does worse than no control, since that is diagnostic. Attitude error RMS is reported alongside, and is the only meaningful metric on the voluntary-motion-only scenario, where suppression is undefined. Every variant is run over identical scenarios with identical sensor seeds, so the comparison is a controlled experiment rather than a collection of separately tuned runs.

Three of the twenty scenarios demand more of the actuator than it can supply, judged by an inverse-dynamics bound that assumes perfect knowledge and exact tracking: case 5 (10 Hz, 3°) needs 7.10 V of a 7.4 V bus; case 14 (8 Hz, 5°) needs 7.50 V; case 20 (20 Hz, 1°) needs 1.84 A against a 1.5 A limit, 9.75 V, and 5.28 W against a 4.4 W continuous rating. No controller could have succeeded on these, so scoring them measures the actuator rather than the algorithm; they are excluded from the headline mean, and all three averages — all scored scenarios, feasible scenarios only, and feasible scenarios that also stayed within limits — are always reported together so the exclusion is visible rather than silent.

D. Controller Comparison on Synthetic Tremor

Table IV gives the six-variant comparison. The variants differ only in the estimator and in whether the sEMG stream is used; plant, scenarios and seeds are identical throughout.

Variant	All (%)	Feas. (%)	Worst (%)	Err (°)	≥90 %
Fixed notch (original)	64.19	90.23	51.51	0.45	15/16
First EKF (defect)	68.75	88.96	30.75	1.23	11/16
Corrected EKF, IMU only	71.03	92.12	85.88	0.20	11/16
EKF, IMU + 2nd oscillator	71.04	92.13	85.90	0.20	11/16
Ablation: no sEMG	65.62	85.18	−41.41	3.04	15/16
sEMG + IMU (delivered)	72.63	93.04	85.31	0.20	15/16

TABLE IV. Controller Comparison, 20 Scenarios

The delivered controller averages 93.04% tremor suppression over the sixteen feasible scenarios and meets the 90% design target on fifteen of them, while holding voluntary-motion error to 0.207° against 0.458° for the original controller. This is the result the controller was designed and tuned against, and it is the figure that justified the architecture: it is the first variant that is not a trade-off, matching the original controller on plain tremor, matching the IMU-only estimator on deliberate movement, and beating both on multi-frequency tremor (Fig. 3). Two properties of this result matter as much as the headline. First, it is not purchased with actuator effort: peak current across the whole sweep is 1.4320 A for the delivered controller against 1.4305 A for the IMU-only estimator, and peak mean coil power 5.719 W against 5.723 W — differences of about 0.1%, comfortably inside the thermal and battery budget. Second, the gain is concentrated where the original controller was weakest: on the scenario combining voluntary movement with tremor it collapses to 51.51%, while the delivered controller holds 85.31%.

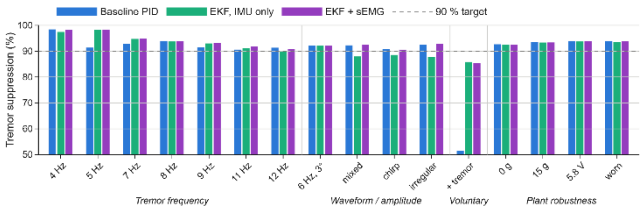


Fig. 3. Tremor suppression across the sixteen feasible scenarios, grouped by frequency, waveform, voluntary motion, and robustness. Dashed line: 90% target.

E. Attribution: the Sensor, not the Mechanism

sEMG enters the estimator as a per-axis, per-sample scaling of the process noise assigned to the voluntary-motion state — in effect a confidence signal deciding how freely the estimator may attribute motion to intent. A reasonable objection is that the benefit comes from that scaling and not from the muscle signal at all. The ablation variant in Table IV exists to answer it: it applies the identical scaling with the confidence held constant and no muscle signal present.

Measure	No sEMG	sEMG+IMU	Sensor gain
Feasible-case mean (%)	85.18	93.04	+7.86 pts
Movement + tremor (%)	−41.41	85.31	+126.7 pts
Voluntary-only error (°)	3.0418	0.2070	14.7× lower

TABLE V. Ablation — What the Sensor Contributes

The ablation behaves as predicted: holding the confidence at a constant favourable value improves multi-frequency tremor suppression and destroys everything involving deliberate movement. Any gain the delivered controller shows over both the IMU-only estimator and this ablation is therefore attributable to the sEMG stream and to nothing else. Inspecting the estimator confirms the designed mechanism directly: tremor mis-attributed to the voluntary state falls by 2.6× to 3.1× on the multi-frequency scenarios (0.171°→0.066°, 0.170°→0.055°, 0.038°→0.014°) with no loss of movement tracking.

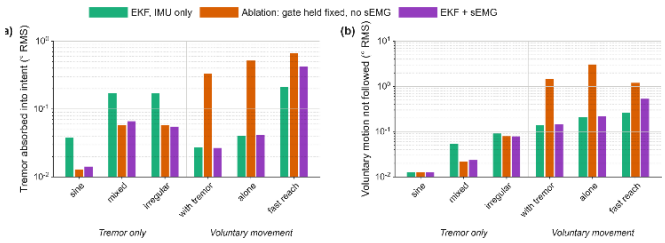


Fig. 4. The sEMG ablation. (a) Tremor mis-attributed to voluntary motion. (b) Voluntary motion missed. The ablation trades (a) for (b) with no muscle signal present — the trade sEMG exists to break. Log ordinates; lower is better.

F. Limitations Identified by Testing

Three targeted tests each found a failure mode. Fast voluntary movement: a 0.8 s reach added to tremor shows the gate at 407 ms against a 350 ms target, costing 2.4× the tracking error of the IMU-only estimator (0.697° vs. 0.289°) for only ≈0.6 points of suppression — a scenario absent from the Table IV benchmark. Rate-of-change detector: an earlier proposal to swap the amplitude detector for a rate-of-change one was tested and retracted after measurement showed it separates the two classes backwards (0.13:1–0.69:1 against a 4:1 requirement). Real forearm sEMG: on two GRABMyo [13] participants, wrist flexion separates as designed (10.1:1, 5.1:1) but extension does not, giving a pooled worst case of 0.1:1–0.8:1 against the 6.3:1 synthetic figure and 4:1 requirement, from a resting DC offset and genuine co-activation absent from the synthetic model.

G. Feasibility Study: Recorded Human Tremor

Sections V-D and V-E used only synthetic tremor (sines, chirps; crest factor 1.414). This study replaces the tremor input with 60 recordings from 14 Parkinson's patients in the public PhysioNet tremordb database [14], [15] (index-fingertip velocity, 100 Hz, checksum-verified), band-pass filtered, integrated to displacement, and RMS-matched to the equivalent synthetic sine so tremor energy is comparable; the sEMG stream remains synthetic throughout, since no public dataset pairs muscle activity with measured tremor. The recordings are markedly less regular than the synthetic input (median crest factor 2.77, range 1.74-6.26, vs. 1.414) and cover a broader dominant-frequency range (median 4.48 Hz, 2.41-8.41 Hz) than the commanded synthetic frequency.

Results are reported over the 46 records inside the 4–20 Hz design envelope (the other 14 are low-amplitude, non-tremor recordings).

Variant	Mean(%)	Worst(%)	≥90%	RMS(°)
Fixed notch chain	81.62	52.65	8/46	0.2080
EKF, IMU only	73.16	29.58	8/46	0.3073
EKF + sEMG	86.81	62.62	21/46	0.1464

TABLE VII. Suppression on Recorded Tremor, 46 In-Band Records

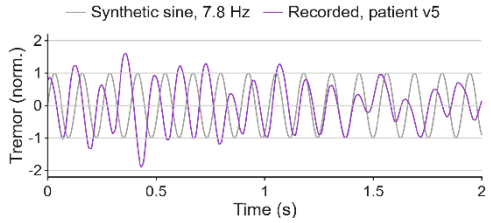


Fig. 5. Recorded human tremor: a conditioned tremordb recording against the RMS-matched synthetic sine at the same dominant frequency.

Two findings stand out. The synthetic benchmark is optimistic: suppression falls from 93.04% (Table IV) to 86.81% on recorded tremor, meeting the 90% target on fewer than half the records. The shortfall is the estimator's, not the plant's or the mechanism's — against a paired synthetic control at the same frequency, the IMU-only estimator loses 23.79 points on recorded tremor while the fixed-notch baseline loses essentially none — not because it performs well, but because its fixed notches never track the input's actual frequency, so its score is largely insensitive to which signal is used. A static, single-frequency filter mis-fits the wandering frequency of real tremor by construction (Section V-E); its apparent robustness reflects that it never adapts, not that it handles recorded tremor well. sEMG narrows that loss to 9.54 points, so the sensor's margin over IMU-only nearly triples on recorded tremor relative to synthetic (+0.92 to +13.65 points). This is a real-signal robustness test of the algorithm, not a clinical result: the muscle signal stays synthetic, the recordings are Parkinsonian fingertip tremor (not wrist tremor during feeding), and a single axis drives both joints.

H. Threats to Validity and Reproduction

In order of impact on these conclusions: no hardware measurement was taken, so every actuator/flexure parameter is unverified; the sEMG signal is synthetic throughout, so every sEMG figure is an upper bound; the plant omits hysteresis, creep and backlash; the 40 ms electromechanical delay the gate's timing rests on is a literature value [12], not measured; and a sustained isometric hold would read as deliberate movement, untested but the safer failure direction. A short synchronised recording of forearm sEMG and limb motion (rest, tremor, reaching) would close most of these at once and is the natural next step before any hardware or clinical claim.

VI. Discussion and Conlcusion

SensoSpoon combines a flexure-based two-axis gimbal, dual voice-coil actuators, dual IMUs and a 13-state EKF with an sEMG intent gate to actively suppress hand tremor during eating. The problem is structural, not a matter of tuning: intent and tremor overlap in frequency, so an estimator that sees only their sum cannot separate them from attitude alone, and the thirteen-state EKF resolves this only partially, which is why the sEMG gate exists. Section V shows containment works: the delivered controller is the first variant that is not a trade-off, with the ablation attributing the gain specifically to the sEMG stream. Across a 20-scenario benchmark it reached 93.04% mean tremor suppression, falling to 86.81% when validated against 60 recorded Parkinsonian tremor traces - the synthetic figure is accordingly an upper bound, the recorded-tremor figure the more realistic estimate. Two limitations emerged from testing: a 2.4× tracking-error cost during fast reaches, and incomplete separation of contraction directions in real forearm sEMG. These results establish technical feasibility, not hardware or clinical validation; the priority next step is a synchronised sEMG/limb-motion recording, alongside rod-end validation, PCB-protection circuitry, sealing verification, and bench control-loop identification.

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